

FORCE 2020 geological penalty matrix (argmax decoding ignores it)

True class

Shale	0	3.5	2.075	3.5	3.75	2	3.5	4	4	3.25	2.75	3
Sandstone	3.5	0	2	3.5	3.5	3	3.75	4	4	3.875	2.5	3.25
Sandstone/Shale	2.375	2	0	3.75	3.75	2.75	4	3.875	4	3.75	3	3
Limestone	3.5	3.5	3.75	0	1.375	2.025	4	3.75	4	3.5	3.625	
Chalk	3.75	3.5	3.75	1.375	0	2.25	2.875	4	3.75	4	3.125	3.75
Marl	2	3	2.75	2	2.25	0	2.5	4	4	3.75	3.375	3.25
Dolomite	3.5	3.75	4	2.625	2.875	2.5	0	3.75	3.25	4	3	3.625
Halite	4	4	3.875	4	4	4	3.75	0	2.75	3.75	3.75	4
Anhydrite	4	4	4	3.75	3.75	4	3.25	2.75	0	4	4	3.875
Coal	3.25	3.875	3.75	4	4	3.75	4	3.75	4	0	2.5	4
Tuff	2.75	2.5	3	3.5	3.125	3.375	3	3.75	4	2.5	0	3.25
Basement	3	3.25	3	3.625	3.75	3.25	3.625	4	3.875	4	3.25	0

clastic

carbonate-marl

evaporite

Predicted class